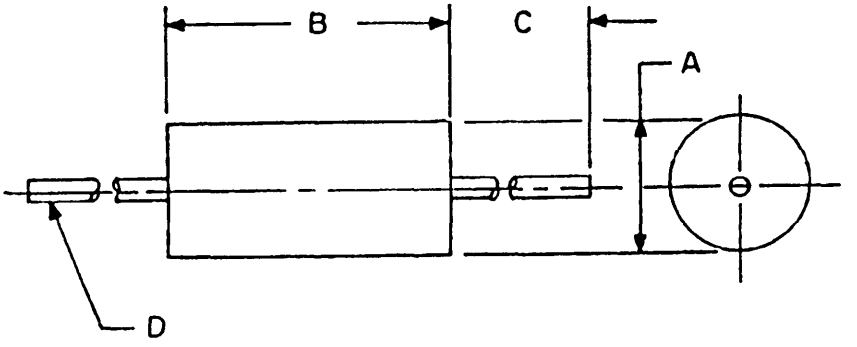
MILITARY SPECIFICATION SHEET

# COILS, FIXED, RADIOFREQUENCY, MOLDED, MICRO-MINIATURE, MAGNETICALLY SHIELDED (FERRITE CORE-FERRITE SLEEVE), ESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-39010.




| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| A | .085 (2.16) | .105 (2.67) |
| B | .240 (6.10) | .260 (6.60) |
| C | 1.250(31.75) | 1.626(41.30) |
| D | .0185 (.47) | .0215 (.55) |

NOTE: Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.


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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Material: Ferrite core - ferrite sleeve.

Weight: 0.30 gram maximum.

Operating temperature range: -55°C to +125°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 300 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Inductance tolerance: See table I.

Q values: See table I.

Self-resonant frequency: See table I.

Percent coupling: 5 percent maximum.

DC resistance: See table I.

Temperature rise: 35°C.

Terminal pull: 5 pounds.

Part number: M39010/12 (dash number from table I).



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TABLE I. Electrical characteristics (initial) and dash numbers.

| Dash number 1/ | Inductance µH | Inductance tolerance ± percent | Q min | Test frequency (MHz) | Self-resonant frequency min (MHz) | DC resistance (25°C) max (ohms) | Rated dc current (mA) 2/ | Incremental current (mA) |
|---|---|---|---|---|---|---|---|---|
| B110** | 11.0 | 5 | 45 | 2.5 | 40 | 2.0 | 250 | 230 |
| B120** | 12.0 | 5, 10 | 45 | 2.5 | 40 | 2.0 | 250 | 230 |
| B130** | 13.0 | 5 | 45 | 2.5 | 36 | 2.5 | 230 | 180 |
| B150** | 15.0 | 5, 10 | 45 | 2.5 | 36 | 2.5 | 230 | 180 |
| B160** | 16.0 | 5 | 45 | 2.5 | 31 | 3.5 | 195 | 150 |
| B180** | 18.0 | 5, 10 | 45 | 2.5 | 31 | 3.5 | 195 | 150 |
| B200** | 20.0 | 5 | 45 | 2.5 | 28 | 4.5 | 170 | 135 |
| B220** | 22.0 | 5, 10 | 45 | 2.5 | 28 | 4.5 | 170 | 135 |
| B240** | 24.0 | 5 | 48 | 2.5 | 25 | 5.0 | 160 | 115 |
| B270** | 27.0 | 5, 10 | 48 | 2.5 | 25 | 5.0 | 160 | 115 |
| B300** | 30.0 | 5 | 48 | 2.5 | 23 | 5.5 | 150 | 105 |
| B330** | 33.0 | 5, 10 | 48 | 2.5 | 23 | 5.5 | 150 | 105 |
| B360** | 36.0 | 5 | 48 | 2.5 | 22 | 7.0 | 140 | 100 |
| B390** | 39.0 | 5, 10 | 48 | 2.5 | 22 | 7.0 | 140 | 100 |
| B430** | 43.0 | 5 | 48 | 2.5 | 21 | 7.5 | 135 | 95 |
| B470** | 47.0 | 5, 10 | 50 | 2.5 | 21 | 7.5 | 135 | 95 |
| B510** | 51.0 | 5 | 50 | 2.5 | 19 | 8.0 | 130 | 90 |
| B560** | 56.0 | 5, 10 | 52 | 2.5 | 19 | 8.0 | 130 | 90 |
| B620** | 62.0 | 5 | 52 | 2.5 | 18 | 9.0 | 120 | 83 |
| B680** | 68.0 | 5, 10 | 55 | 2.5 | 18 | 9.0 | 120 | 83 |
| B750** | 75.0 | 5 | 55 | 2.5 | 16 | 14.0 | 95 | 70 |
| B820** | 82.0 | 5, 10 | 55 | 2.5 | 16 | 14.0 | 95 | 70 |
| B910** | 91.0 | 5 | 55 | 2.5 | 14 | 16.0 | 90 | 65 |
| B101** | 100.0 | 5, 10 | 55 | 2.5 | 14 | 16.0 | 90 | 65 |

See footnotes at end of table.



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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

| Dash number 1/ | Inductance µH | Inductance tolerance ± percent | Q min | Test frequency (MHz) | Self-resonant frequency min (MHz) | DC resistance (25°C) max (ohms) | Rated dc current (mA) 2/ | Incremental current (mA) |
|---|---|---|---|---|---|---|---|---|
| B111** | 110.0 | 5 | 34 | .79 | 9.5 | 10 | 90 | 40 |
| B121** | 120.0 | 5, 10 | 34 | .79 | 9.5 | 10 | 115 | 40 |
| B131** | 130.0 | 5 | 34 | .79 | 8.8 | 12 | 105 | 35 |
| B151** | 150.0 | 5, 10 | 34 | .79 | 8.8 | 12 | 105 | 35 |
| B161** | 160.0 | 5 | 34 | .79 | 8.1 | 15 | 90 | 32 |
| B181** | 180.0 | 5, 10 | 34 | .79 | 8.1 | 15 | 90 | 32 |
| B201** | 200.0 | 5 | 34 | .79 | 7.8 | 20 | 80 | 30 |
| B221** | 220.0 | 5, 10 | 36 | .79 | 7.8 | 20 | 80 | 30 |
| B241** | 240.0 | 5 | 36 | .79 | 7.3 | 23 | 75 | 27 |
| B271** | 270.0 | 5, 10 | 36 | .79 | 7.3 | 23 | 75 | 27 |
| B301** | 300.0 | 5 | 36 | .79 | 6.8 | 26 | 70 | 25 |
| B331** | 330.0 | 5, 10 | 36 | .79 | 6.8 | 26 | 70 | 25 |
| B361** | 360.0 | 5 | 36 | .79 | 6.0 | 35 | 65 | 23 |
| B391** | 390.0 | 5, 10 | 36 | .79 | 6.0 | 35 | 65 | 23 |
| B431** | 430.0 | 5 | 36 | .79 | 5.6 | 37 | 60 | 21 |
| B471** | 470.0 | 5, 10 | 38 | .79 | 5.6 | 37 | 60 | 21 |
| B511** | 510.0 | 5 | 38 | .79 | 5.0 | 40 | 58 | 20 |
| B561** | 560.0 | 5, 10 | 40 | .79 | 5.0 | 40 | 58 | 20 |
| B621** | 620.0 | 5 | 40 | .79 | 4.3 | 48 | 54 | 18 |
| B681** | 680.0 | 5, 10 | 40 | .79 | 4.3 | 48 | 54 | 18 |
| B751** | 750.0 | 5 | 40 | .79 | 3.8 | 54 | 47 | 17 |
| B821** | 820.0 | 5, 10 | 40 | .79 | 3.8 | 54 | 47 | 17 |
| B911** | 910.0 | 5 | 40 | .79 | 3.5 | 65 | 43 | 15 |
| B102** | 1000.0 | 5, 10 | 40 | .79 | 3.5 | 65 | 43 | 15 |

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the failure rate symbol (e.g., M, P, R, S) and will be added to the end of the dash number.

2/ The rated dc current is based on 90°C ambient temperature with a 35°C rise.



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TABLE II. Electrical characteristics (final).

| Inspection group | Allowable variation from initial measurement | | Allowable % from specified minimum value in electrical characteristics (initial) table | |
|---|---|---|---|---|
| | Inductance | DC resistance | Self-resonant frequency | Q |
| | Percent | | | |
| Qualification inspection | | | | |
| Group II | ±5 | --- | --- | -10 |
| Group IV | ±10 | ±(5%+.001 ohm) | 1/ | -20 |
| Group V | ±5 | ±(2%+.001 ohm) | 1/ | -20 |
| Quality conformance inspection | | | | |
| Group C | | | | |
| Subgroup 1 | ±5 | ±(2%+.001 ohm) | 1/ | -20 |
| Subgroup 3 | ±5 | --- | --- | -10 |
| Subgroup 4 | ±10 | ±(5%+.001 ohm) | 1/ | -20 |

1/ The self-resonant frequency shall be not less than the value specified in table I.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Air Force - 11, 99
DLA - ES

User activities:
Army - ME
Navy - AS, CG, MC, SH
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0657-02)



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